

A Packing of 10 Four-Spaces in $\mathbb{R}^{10}$

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Abstract

We give 10 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.666373. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.639721, obtained by deleting 1 of the 11 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 10 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.666373\dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 10$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.639721, obtained by deleting 1 of the 11 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 10/9 = 2.666667$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N10.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
266637381476214200847209342615355184519832176270148744473398349745746763
699309284361096061739642678435385676779661758289348570227805855970546764
6898859994550504578153530048103789388145532307344
```

denominator:

```
999999999997167770987013650521643420846930166808731751582898676369067421
097765268855468981240296775574683057696974567457982540852997670590494784
376370567137392502058622498917758918344126191551
```

B Coordinates

10 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N10.txt`.

```
-397411530722 -700068176272 9482037364 303354739008 104194217491 -189740531625 477068811009 -309451008857 443168897 -50661119897 278882995280 -706597503087
-576469640916 271012701826 208675385203 -387759884969 766198900 -14322275946 388247439233 -268435014003 17648508303 599659017322 -188614445818 -71408565425
65062437397 329377418595 -463514443076 360930816340 21150634230 -704876419520 -342915324191 -121910303075 27648936 -209996146779 270893173497 56852244585
-190310060573 -65093044546 -84028035134 427489523000 92911474810 -328424880992 -330988716854 -324105938965 172522 94636660216 133786864128 573068022963
38911567042 -94419862432 613446233079 276095493743 -338044266865 17053837844 176434822387 -5338218318 18241159245 351457929006 -375504500800 27549288836
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-113826281505 390154569805 357664400336 58755658693 97780860400 172350814513 215037354582 -39746156754 4221233128217 -41509964407 16110197767 300142080968
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-168755511357 147144500720 61073466786 -8798124501 974249109107 -725079620944 63278069283 348554215248 955102001884 -358271451157 -130450090103 337800101418
-497521678769 -87101775937 -129232416120 -401599937 9903317912 42041858055 -557183052437 -75156147597 -385440490335 -619930707320 91987133345 133586655135
486731253510 444842760118 8041863020 -140991479019 -417806312900 -180628076046 77762805557 -4972958037 7181842640686 51262896196 518896639523 401147177872
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-68164106753 -639207554897 -8463531376 28949748259 724623712055 100774986615 -189691231982 -41964604386 90759266785 -72326079455 -450296873130 -145523775393
-59133927946 -192378902372 -636740711378 -96744671639 2023192718 -180833597838 379973883229 8306169192 21637845566 206035404706 405353096179 54345029451
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-11726076506 -50392249445 -301455062865 1770961085 98898976356 651992794578 263479487814 -14347353022 055532073927 680410711893 -19157268662 -387053570512
179714986532 30461568844 -154537849674 -5793605723 366939649096 469214443158 27980943646 163622720055 33973797031 333174653238 -42237238130 267604410406
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64308607946 402294903781 270662665409 69325494048 -421894796034 -226609862975 99205056741 -749261747 8349830960322 186928017551 6889824825 -544884012276
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-331715150997 256856475026 -245268844430 -41174931 12883715973 335793668882 236546222666 8333913109
```